

Name: NANDYALA SHREYAS REDDY NANDYALA SHREYAS REDDY | **Reg. No:** RA2511003010903

Unit – 2 : Session – 9 : SLO – 1 | Polymorphism Classification

■ Activity – Mystery Boxes (Answers)

- A. area() in both Circle and Rectangle → RUN-TIME POLYMORPHISM (if they share a base class/interface and override area(); resolved by object type at run time)
- B. Printer with print(String) and print(int) → COMPILE-TIME POLYMORPHISM (method overloading — chosen by argument types at compile time)
- C. Loop printing "Hello" five times → NO POLYMORPHISM (plain iteration)
- D. Subclass overrides speak() from superclass → RUN-TIME POLYMORPHISM (overriding)
- E. Method that adds two numbers → NO POLYMORPHISM (ordinary method)
- F. Animal reference pointing to a Dog object calling makeSound() → RUN-TIME POLYMORPHISM (dynamic dispatch through base reference)